

Rahul Gupta

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Senior software engineer with 13+ years of experience building high-performance networking and distributed systems in C/C++ and Go. Experienced in developing scalable product features, implementing secure networking protocols, and mentoring engineers on complex production systems.

TECHNICAL SKILLS

Languages: C, C++, Go, Python

Systems Programming: Multithreading, synchronization, memory management, low-latency systems design

Distributed Systems: High-throughput backend services, scalable service architecture, API design

Debugging & Performance Engineering: gdb, perf, Valgrind, tcpdump, production incident debugging and profiling

Infrastructure & Platform: Linux systems programming, Docker, Kubernetes, CI/CD pipelines

Engineering Practices: System design, architecture reviews, observability, mentoring and technical leadership

PROFESSIONAL EXPERIENCE

Senior Software Engineer | F5, Inc. | Hyderabad, India

May 2020 - Aug 2025

Led development of high-performance Global Server Load Balancing (GSLB) capabilities in F5 BIG-IP DNS, enabling geo-distributed traffic management across data centers handling up to 100M DNS query evaluations/sec.

- Designed and implemented a multi-threaded C++ request-processing pipeline for DNS-based traffic steering, optimizing scheduling, synchronization, and memory locality for deterministic low-latency resolution at scale.
- Built intelligent traffic routing algorithms including weighted round-robin, geo-proximity routing, and health-based failover to ensure high availability and efficient load distribution across regions.
- Leveraged consistent hashing to deterministically route queries to appropriate GTM instances and data centers, improving cache locality and system scalability under high load.
- Integrated secure DNS protocols (DoT, DoH, ODoH) into the execution pipeline, maintaining throughput and latency SLAs under encryption overhead.
- Engineered Go-based asynchronous orchestration services forming the control plane, enabling distributed configuration, policy management, and cross-component communication.
- Diagnosed and resolved complex production issues including race conditions, deadlocks, memory corruption, and tail-latency regressions using advanced debugging and profiling techniques.
- Built Python-based automation and regression framework (~95% coverage) validating correctness, performance, and failover behavior across releases.
- Mentored 5+ engineers on concurrency, distributed systems design, and debugging, influencing architectural decisions across DNS/GSLB subsystems.

- Championed AI-assisted development practices to enhance engineering velocity and code quality across distributed systems

Associate Consultant | GlobalLogic | Gurugram, India

May 2019 - May 2020

Contributed to development of telecom data processing platforms handling large-scale event streams for real-time network analytics and mediation.

- Built high-throughput C++ data mediation services processing millions of call data records (CDRs) per day in distributed processing environments.
- Implemented pipeline optimizations and improved fault tolerance mechanisms, increasing system throughput and stability under heavy workloads.

Technical Leader | Capgemini Engineering (formerly Altran/Aricent) | Gurugram, India

July 2012 – Oct 2018

Contributed to development of telecom billing and Intelligent Network (IN) platforms supporting real-time call and service processing in large-scale telecom deployments.

- Led design and implementation of high-performance C/C++ backend components for telecom billing and IN platforms handling real-time transaction processing.
- Built an internal automation and observability platform adopted across multiple customer deployments in the CALA region, reducing troubleshooting and analysis time by ~90%.
- Developed an IM-SSF simulation service enabling validation of SIP-based IN features without telecom lab infrastructure, accelerating feature development and testing.
- Diagnosed and resolved complex production issues involving concurrency, memory management, and distributed system failures.
- Mentored 10+ engineers on systems debugging, performance profiling, and scalable system design.

ENGINEERING RECOGNITION

- Spotlight of the Month - GlobalLogic | 2019

Recognized for technical ownership and delivery of high-impact backend system features.

- Leadership Recognition - Nokia/Arcent | 2016

Acknowledged for engineering excellence and automation improving system reliability.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Kalinga Institute of Industrial Technology

2012 | CGPA: 8.45